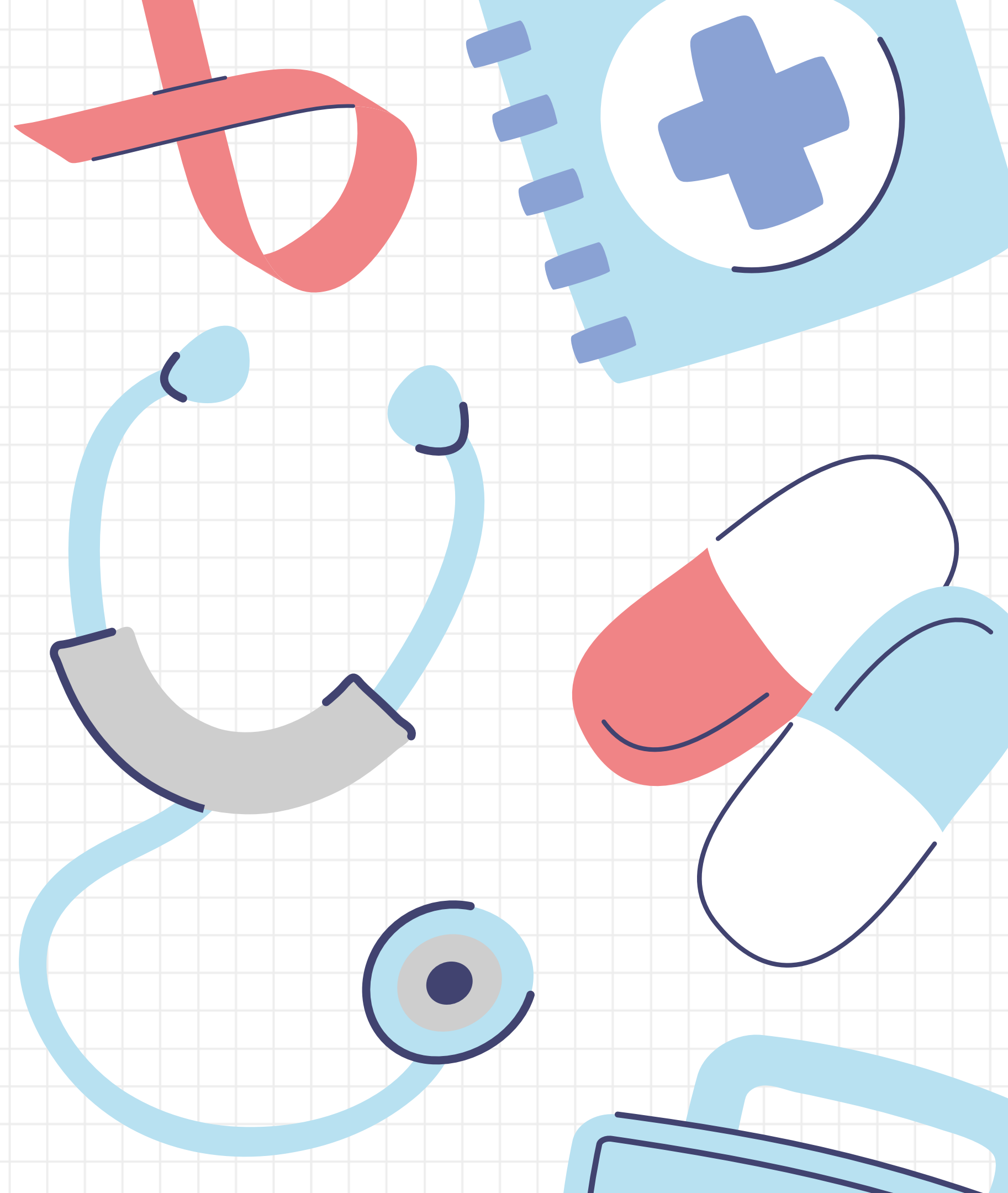
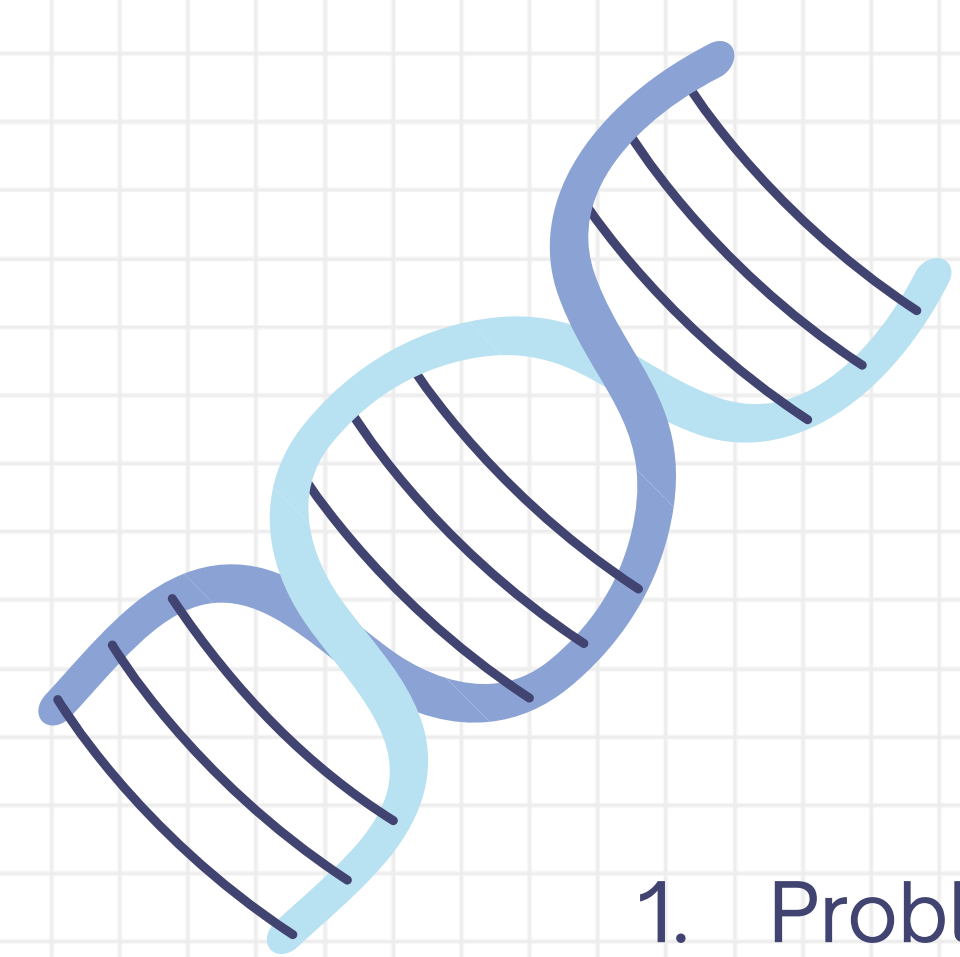


Chin Jia Li

Identifying Skin Cancer





Content

- | | |
|------------------------------|-----------------------------------|
| 1. Problem Statement | 7. Results & Evaluation |
| 2. Data Overview | 8. Project Demo |
| 3. Objective & Key Questions | 9. Measure of Success |
| 4. Methodology | 10. Challenges & Limitations |
| 5. EDA & Key Findings | 11. Future Work & Recommendations |
| 6. Modelling Approach | 12. Tech Stack |



Problem Statement

- Skin lesions are very difficult to tell apart from normal moles / scars on our skin.
- Everyone is affected by this problem, especially those that have high exposures to the sun.
- People can become more aware of skin cancer symptoms with this program and get diagnosed properly by a doctor.



Data Overview

Dataset: <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>

Granularity: Each row contains data on the patient's information (gender, age) and details on the skin lesion (type, diagnosis method, localization)

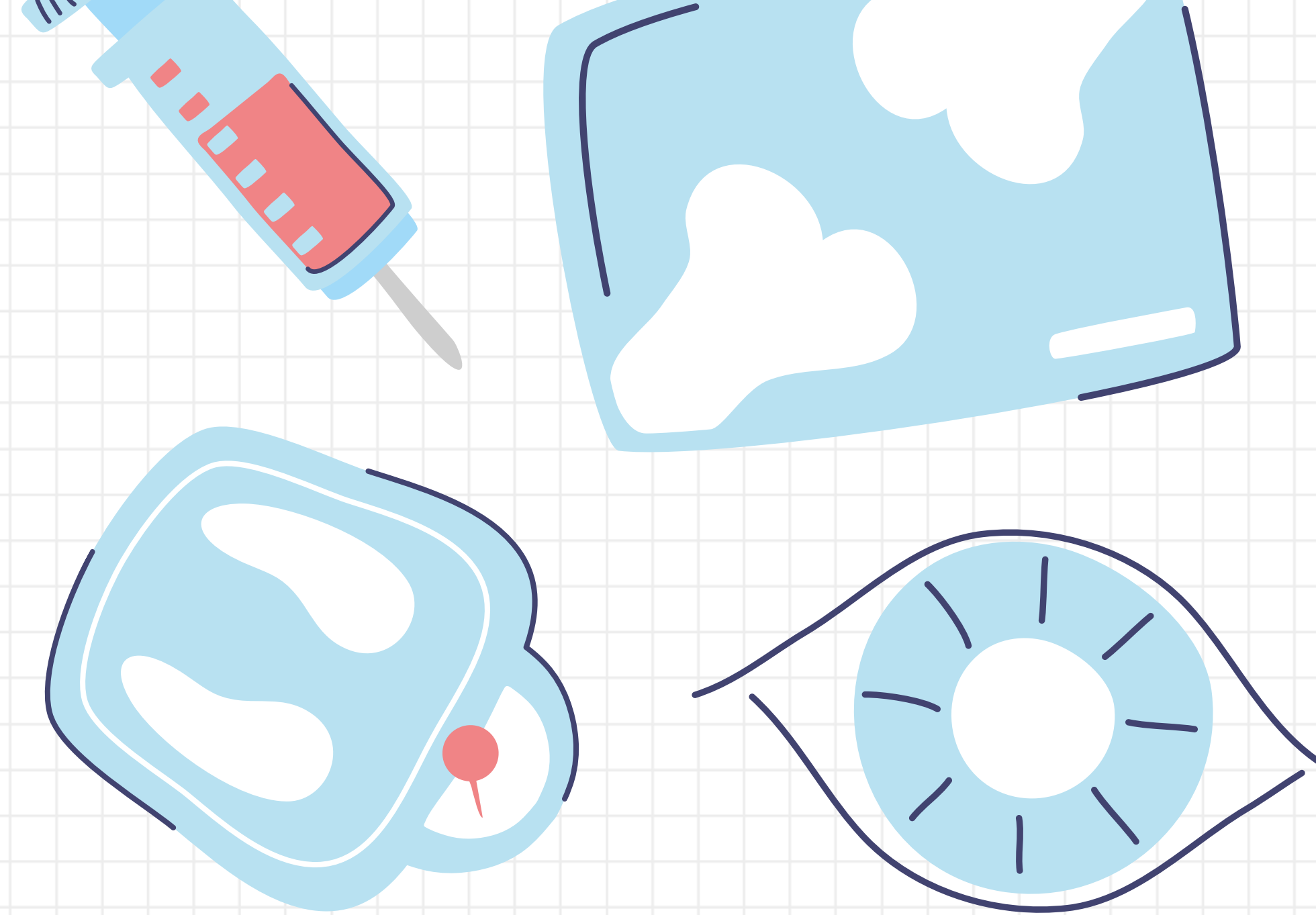
Size: 10015 images

Target Variable: Identify the type of skin cancer from the dataset's 7 types of skin cancer with an image of a skin lesion.

	lesion_id	image_id	dx	dx_type	age	sex	localization	cell_type	image_path
0	HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp	Benign keratosis-like lesions	ISIC_0027419.jpg
1	HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp	Benign keratosis-like lesions	ISIC_0025030.jpg
2	HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp	Benign keratosis-like lesions	ISIC_0026769.jpg
3	HAM_0002730	ISIC_0025661	bkl	histo	80.0	male	scalp	Benign keratosis-like lesions	ISIC_0025661.jpg
4	HAM_0001466	ISIC_0031633	bkl	histo	75.0	male	ear	Benign keratosis-like lesions	ISIC_0031633.jpg

Project Objectives

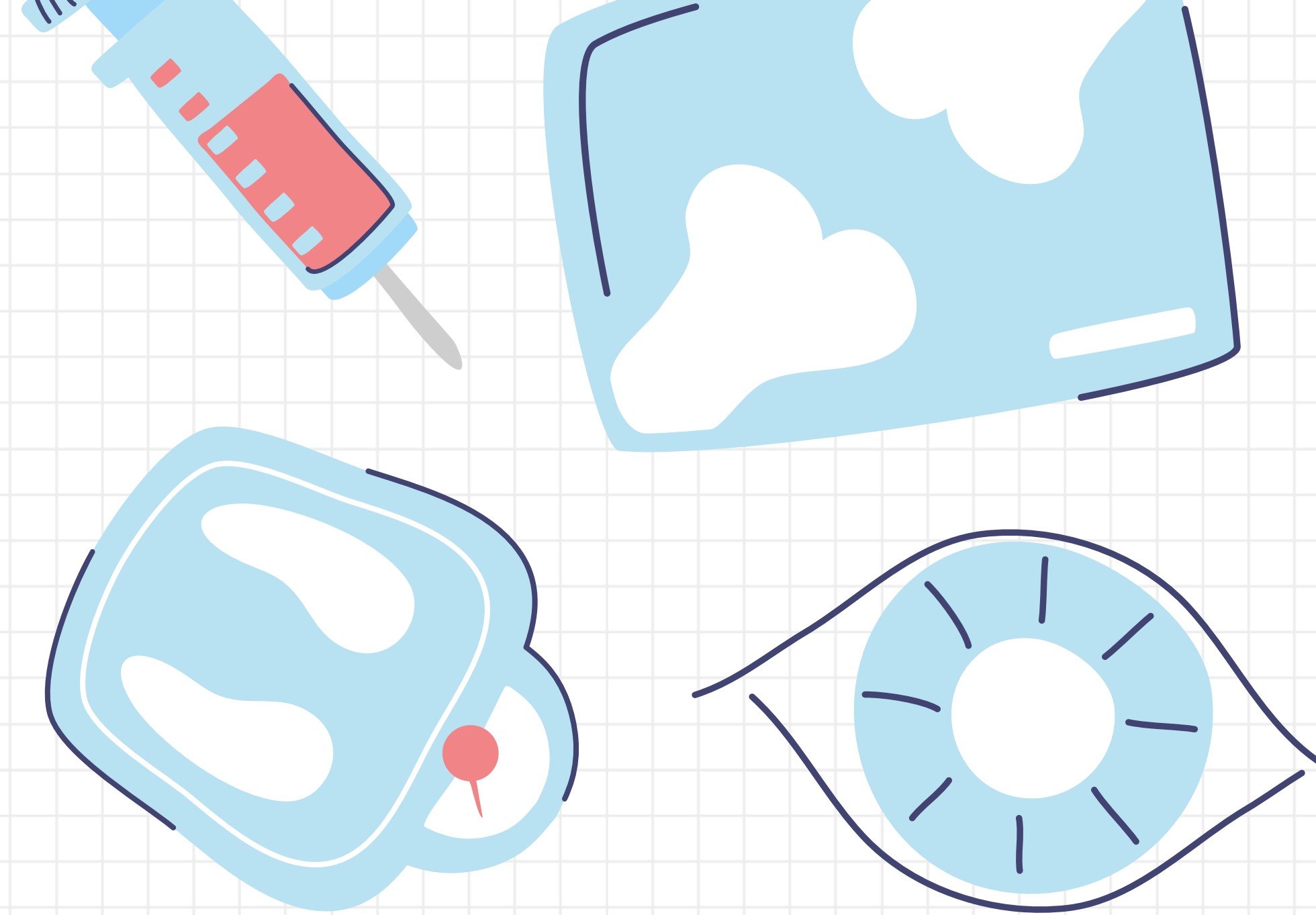
- To develop a model capable of accurately classifying different types of skin cancer with images.
- To evaluate the effectiveness of deep learning models in distinguishing between multiple skin lesion categories.
- To determine clinical relevance by assessing model performance with success metrics such as Confusion Matrix Display and Accuracy Score.



Objective & Key Questions

Key Questions

- How accurately can a deep learning model classify the seven different skin lesion types in the HAM10000 dataset?
- Which skin cancer classes are most frequently misclassified, and what does this indicate about class imbalance, visual similarity between lesions, or model limitations?



Objective & Key Questions

Methodology

Preprocessing:

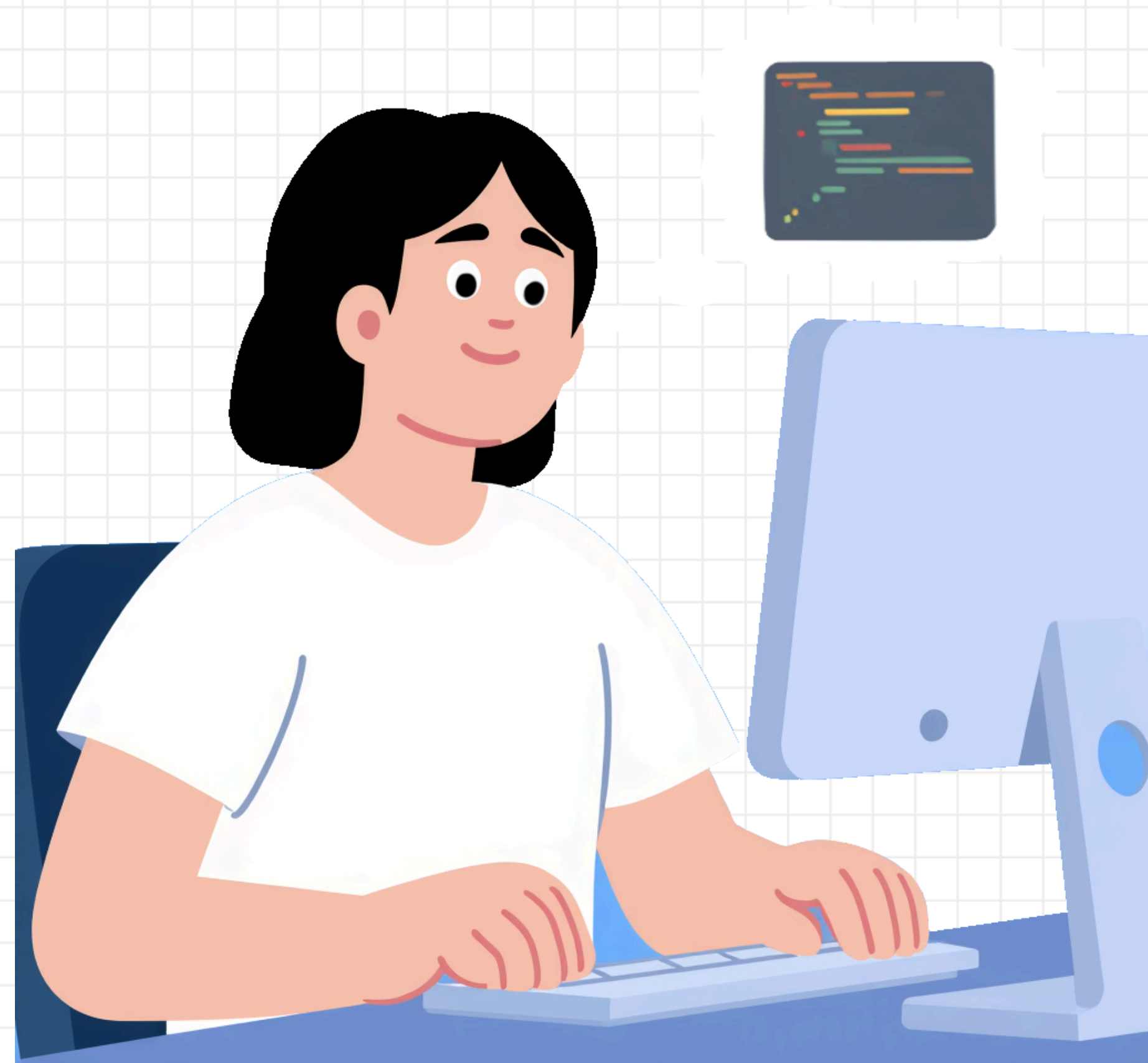
- Cleaned the data by removing invalid values and filling missing values.

Model Training:

- Created data generators with augmentation and generators to train the model with.
- Defined the model's layers, then compiled and fitted the model.

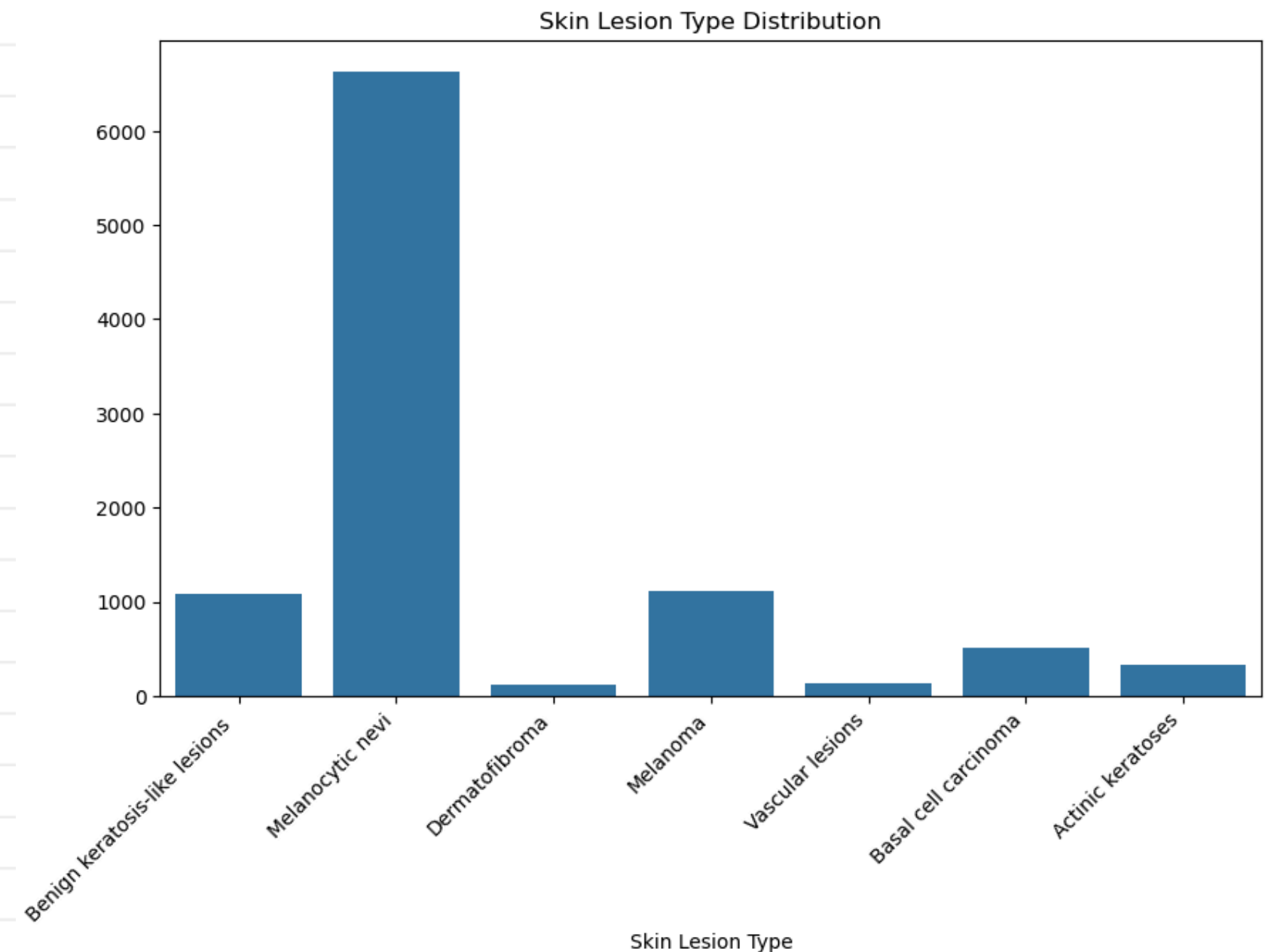
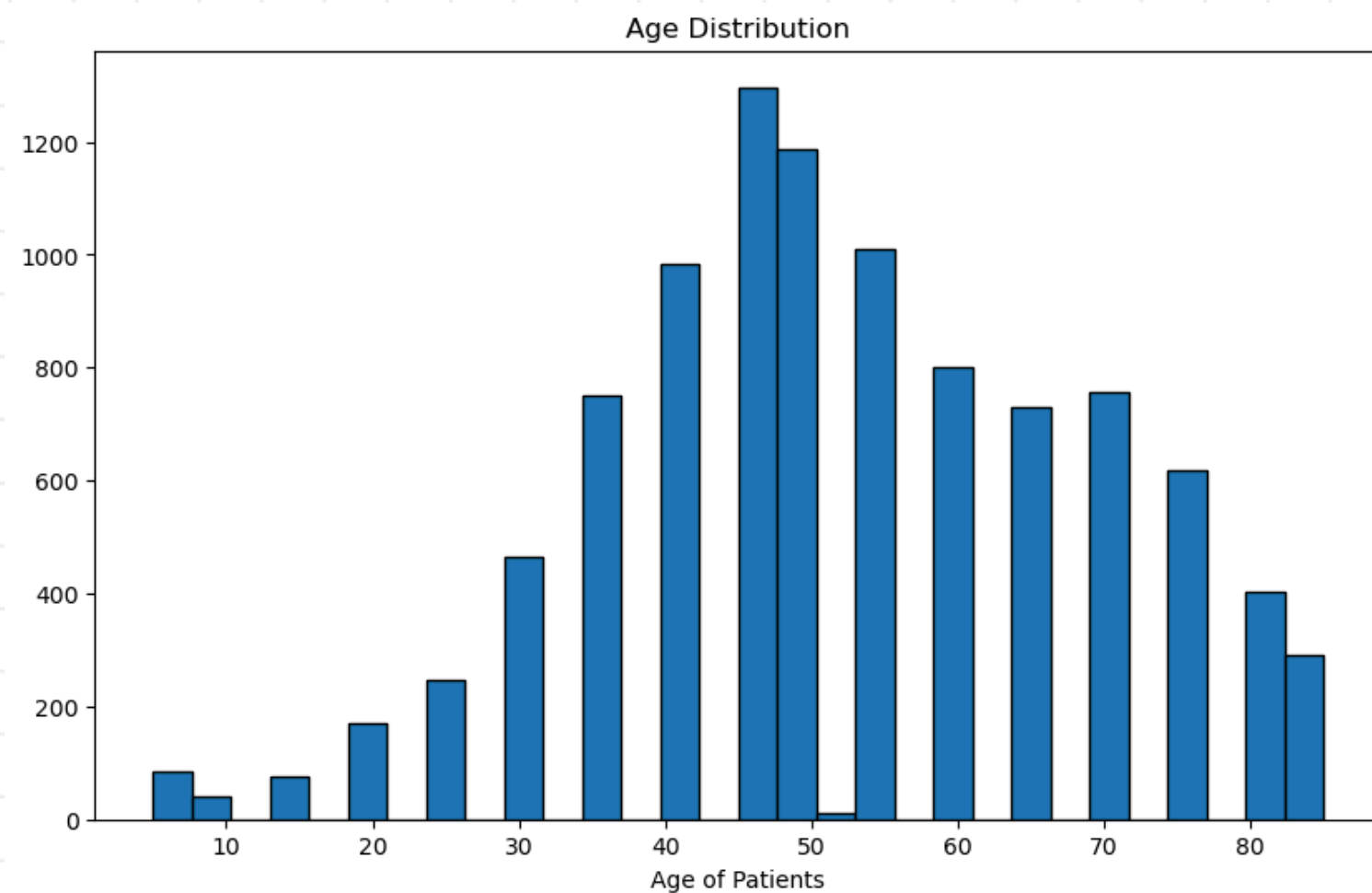
Evaluating:

- Plotted training loss vs validation loss graph
- Plotted Confusion Matrix Display
- Displayed accuracy score of the model



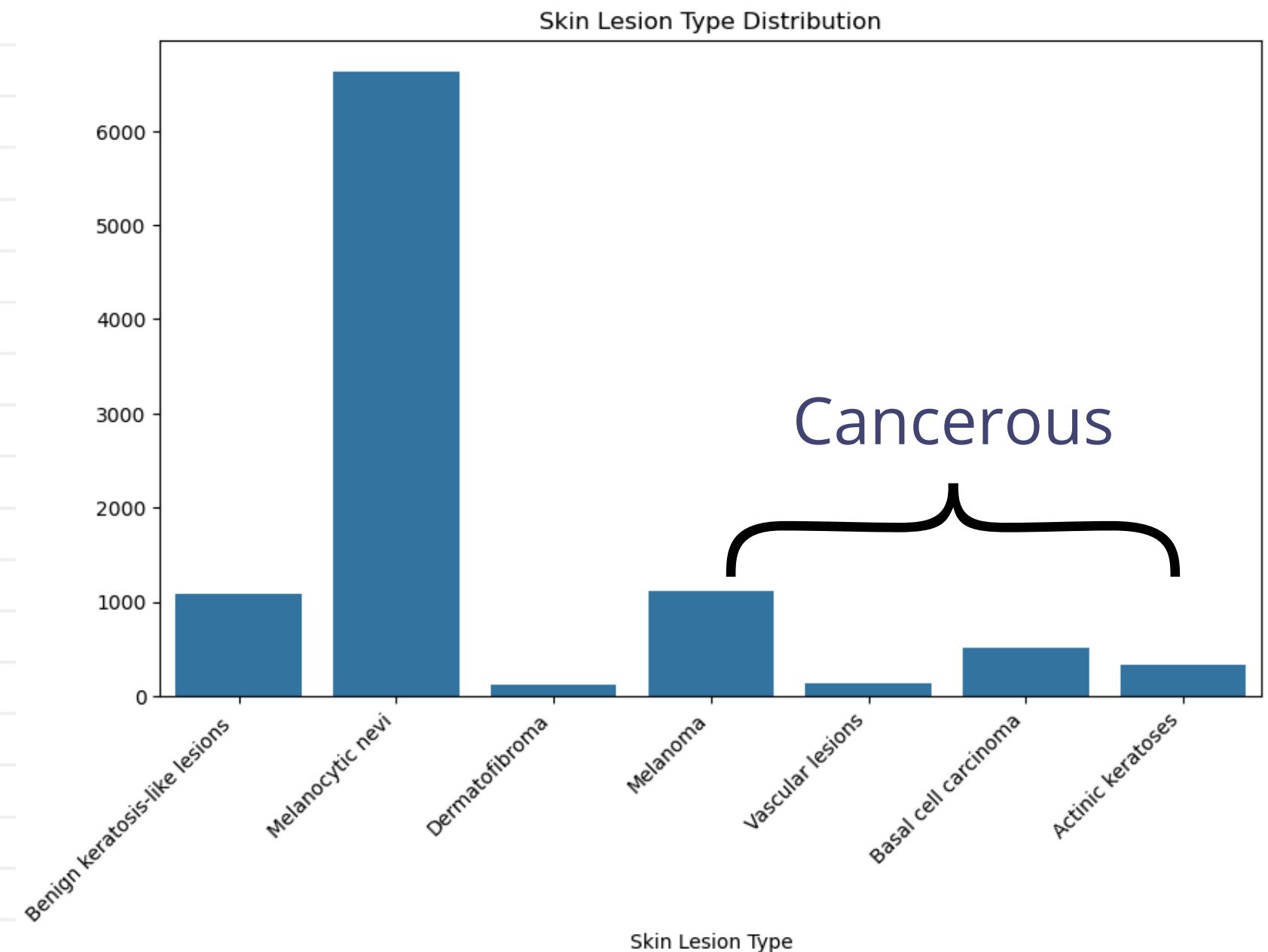
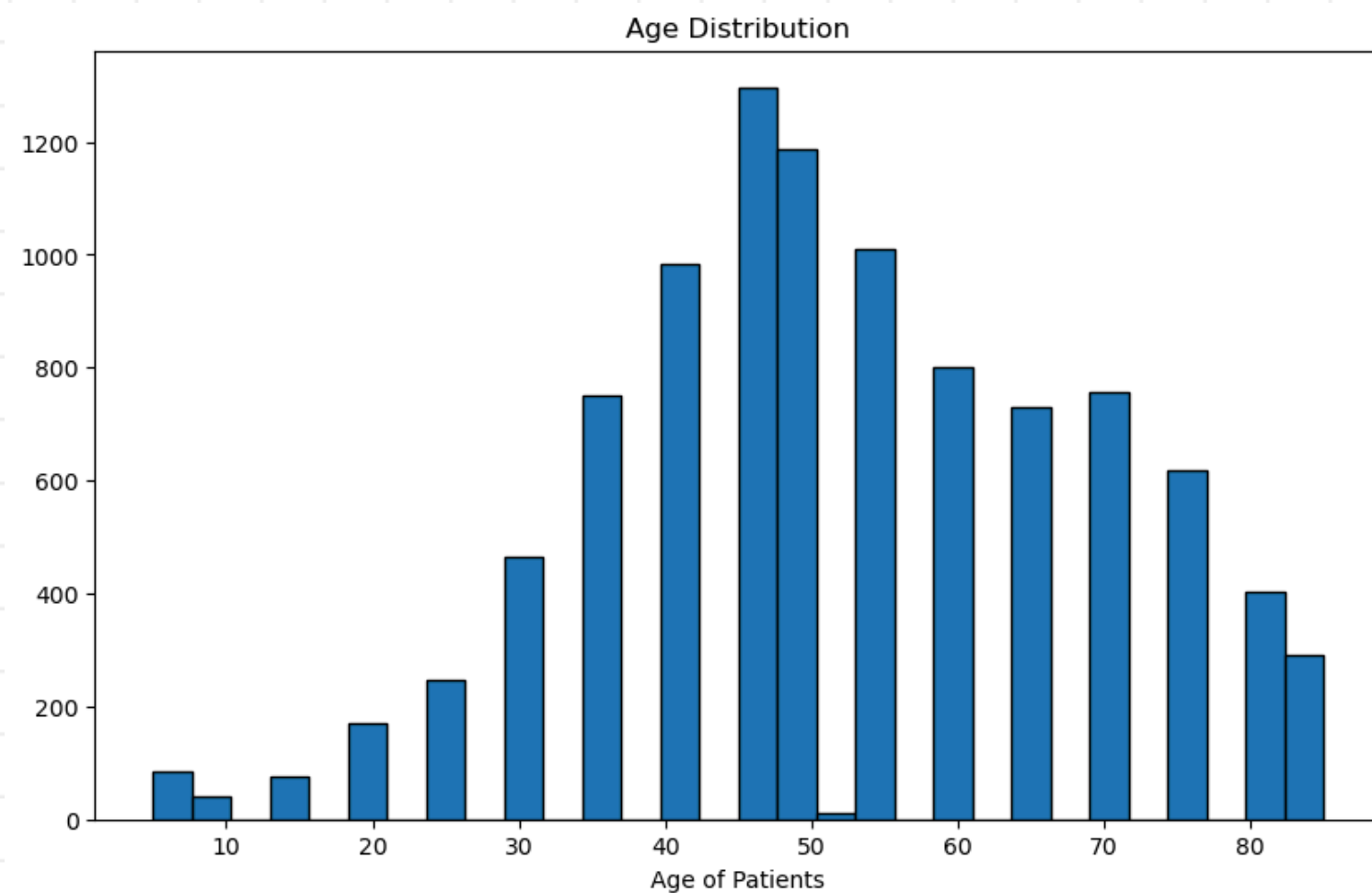
EDA & Key Findings

- There were 57 missing values in the age column, so I filled it with the average age of all patients.
- There are 7 classes of skin lesion types found in the dataset.
- The dataset is shown to have an extreme case of class imbalance for the type of Skin Lesion, with Melanocytic Nevi being the skin lesion type with the most cases.



EDA & Key Findings

- There were 57 missing values in the age column, so I filled it with the average age of all patients.
- There are 7 classes of skin lesion types found in the dataset.
- The dataset is shown to have an extreme case of class imbalance for the type of Skin Lesion, with Melanocytic Nevi being the skin lesion type with the most cases.

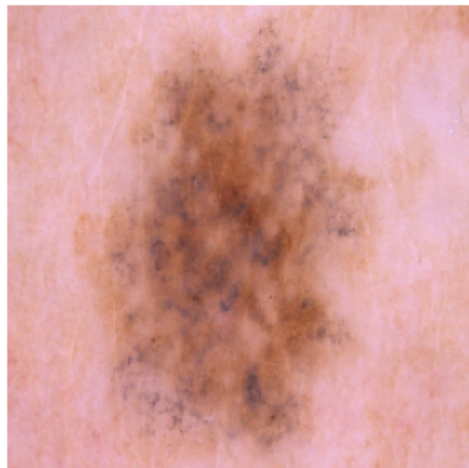


EDA & Key Findings

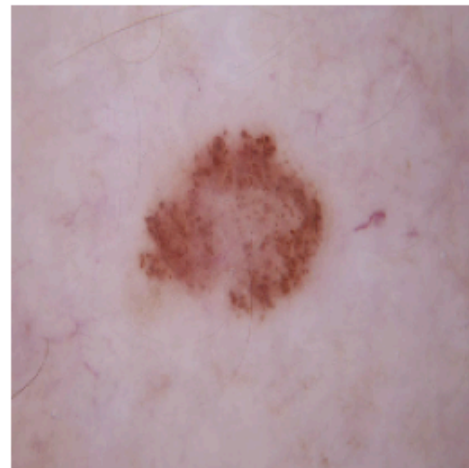
Some sample images of the images in the training folder for the model, and sample augmentation images.

Sample Skin Cancer Images

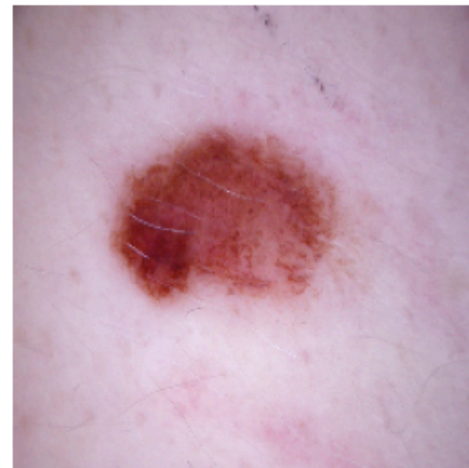
Melanocytic nevi



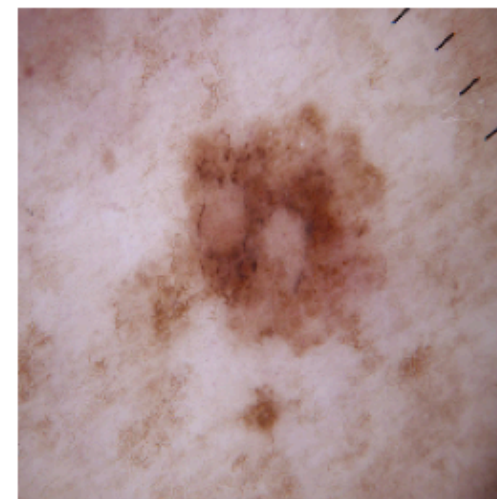
Melanocytic nevi



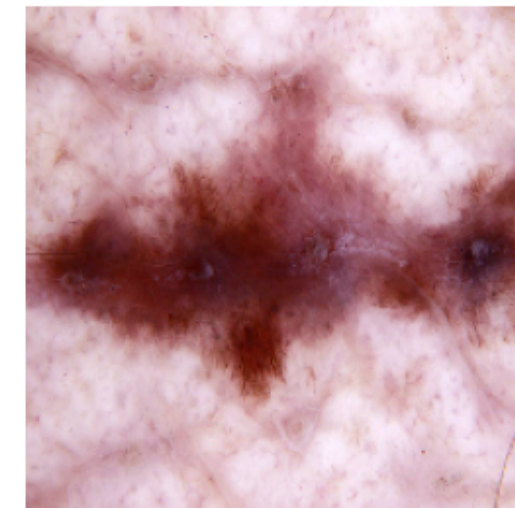
Melanocytic nevi



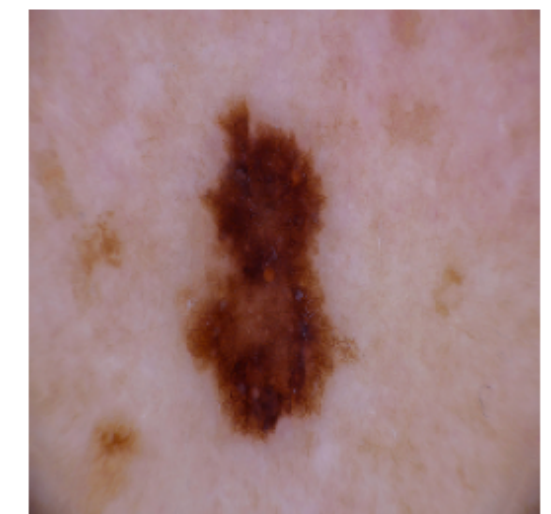
Benign keratosis-like lesions



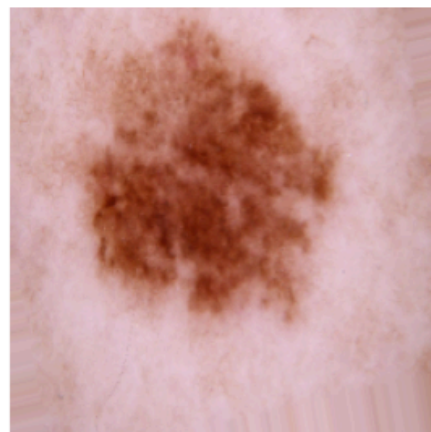
Melanoma



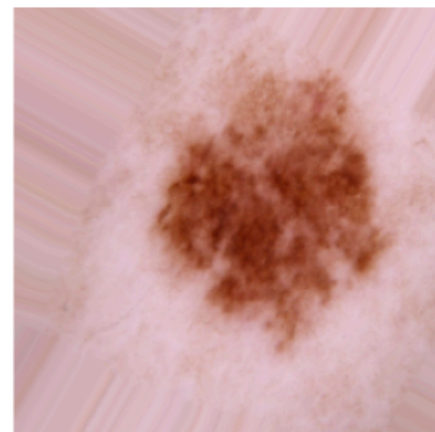
Melanoma



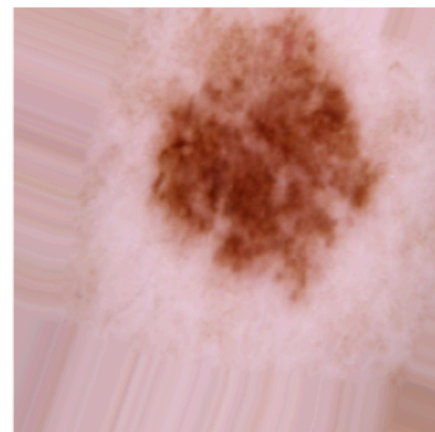
Original



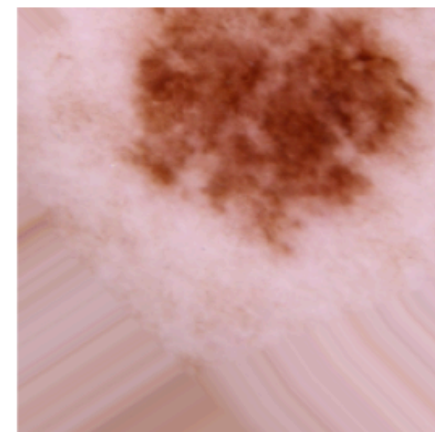
Augmented 1



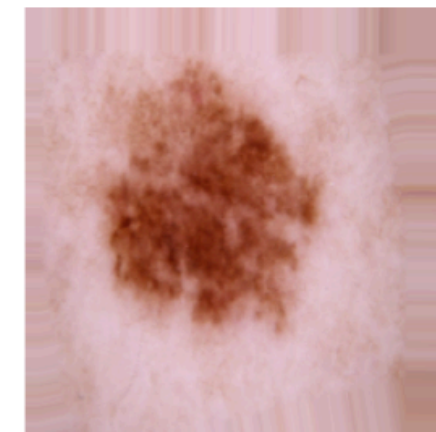
Augmented 2



Augmented 3

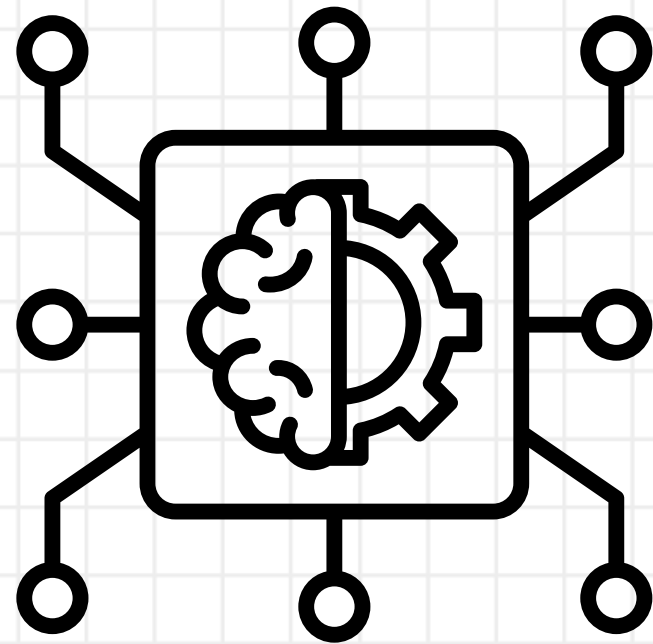


Augmented 4



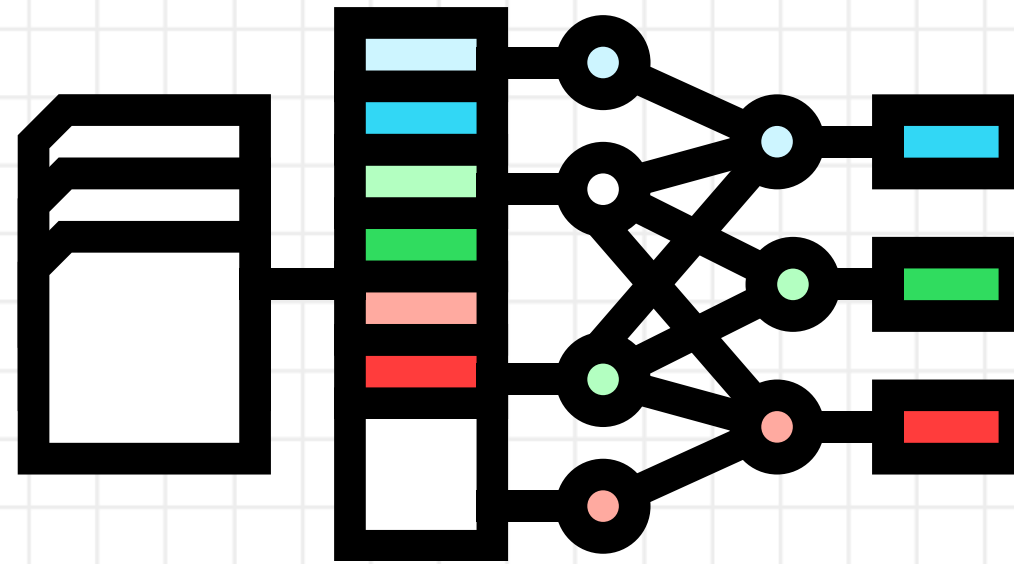
Modelling Approach

I used a 80/20 split by separating 8000 images in the training folder and 2015 images in the testing folder. I encoded the different types of skin lesion types to



Simple CNN

- What are the minimum layers I need to get a good prediction / accuracy?



Deeper CNN

- With more layers in the model, will the accuracy improve?



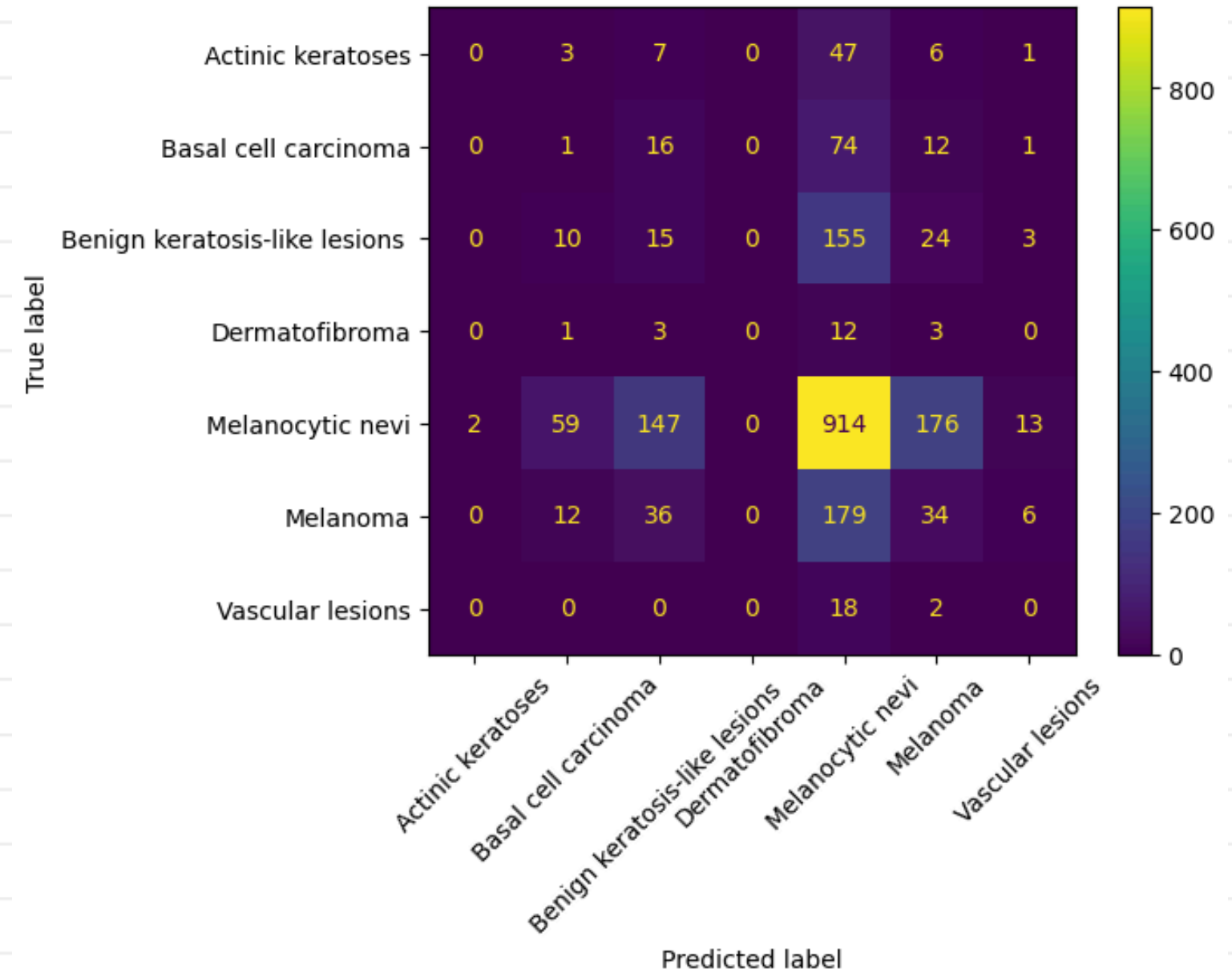
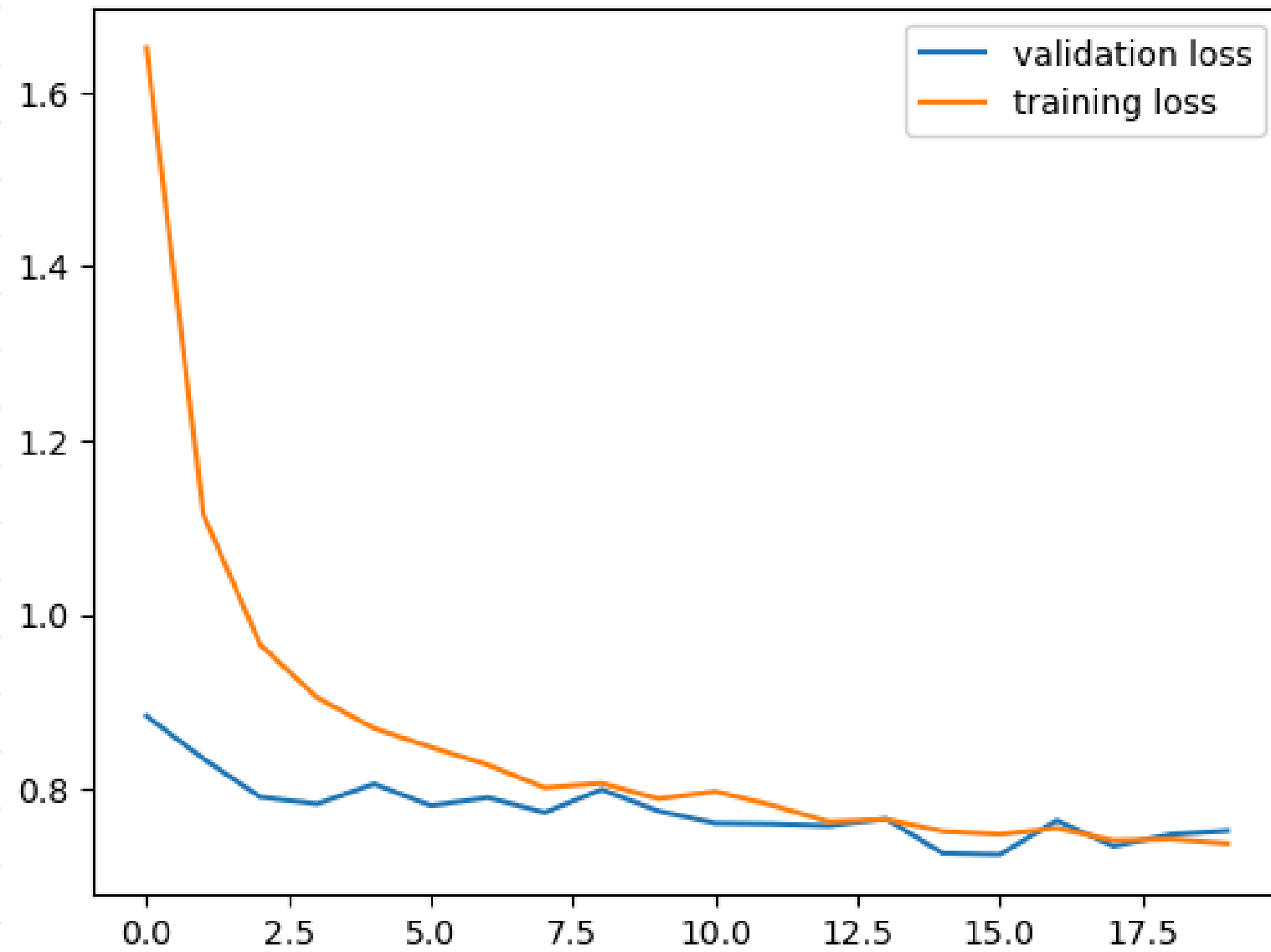
MobileNet v2

Transfer Learning Model

- Pretrained model will take less time to train because it only needs to train the final layer.

Results & Evaluation

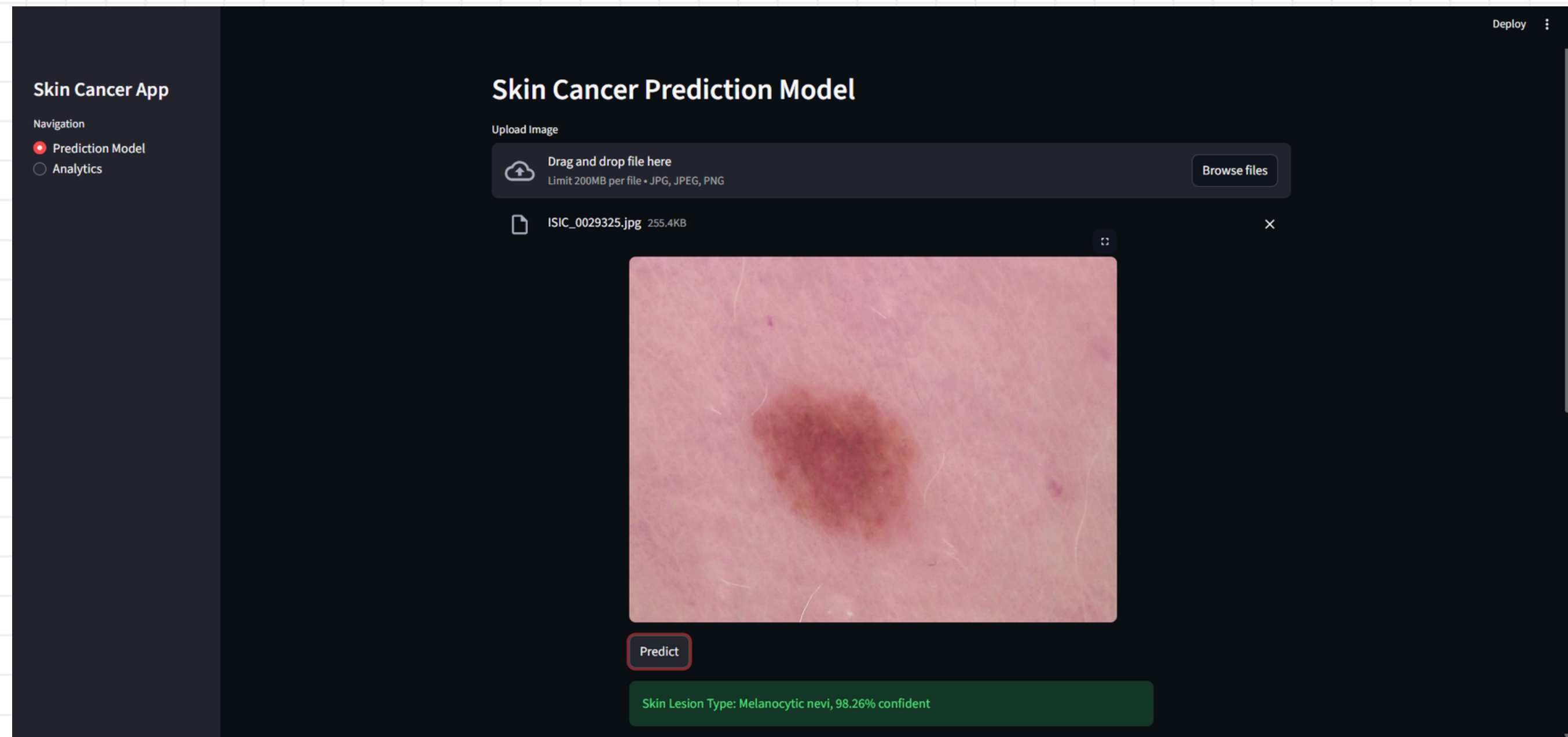
The transfer learning model managed to achieve an accuracy score of 74.75%



This means that the model has a 74% chance of correctly identifying the type of skin lesion from the image.

Project Demo

Link:



Measure of Success

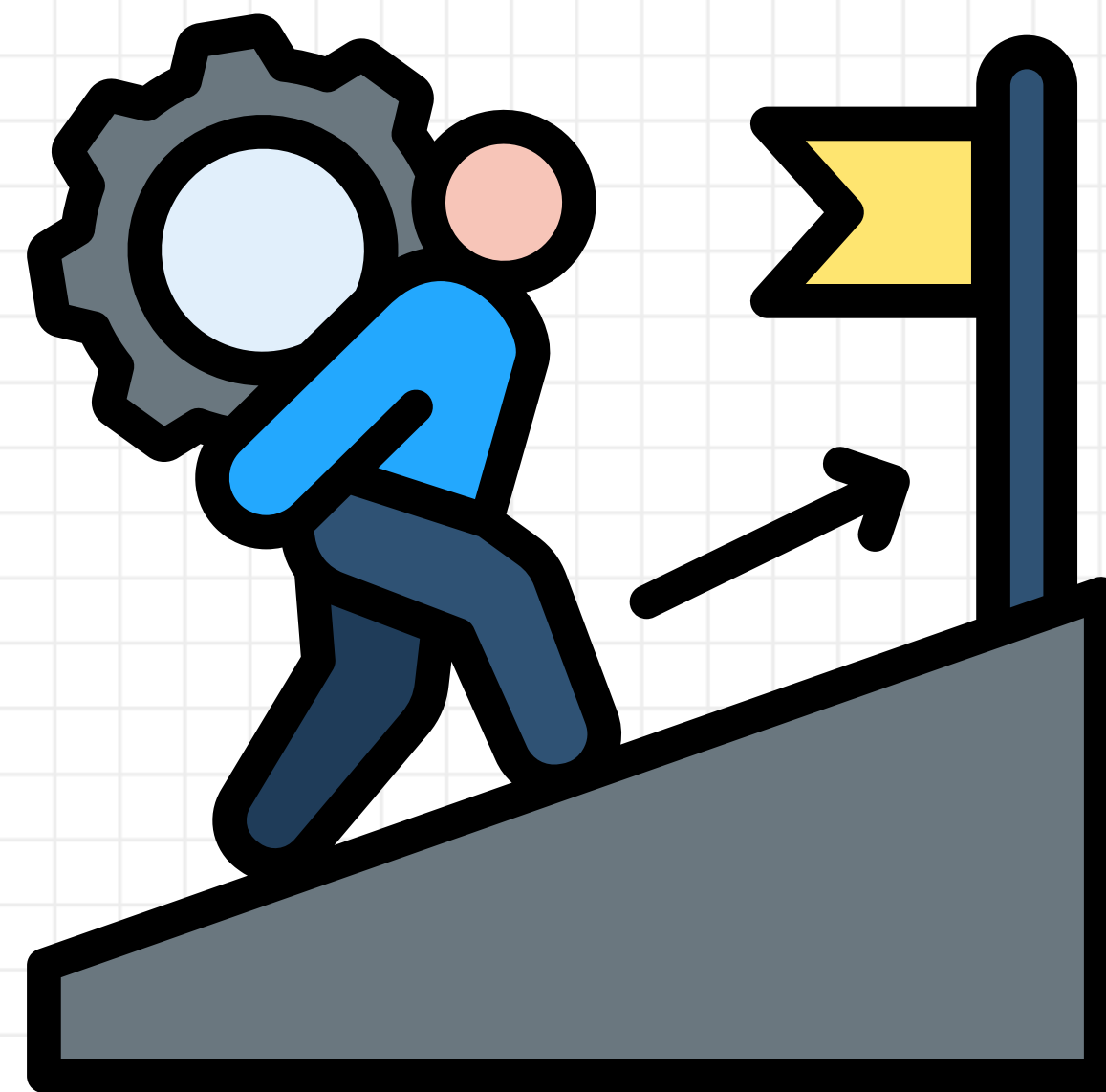
- The model is considered successful if its accuracy is more than 70%.
- The transfer learning model achieved the best accuracy score (74.75%) out of all 3 models.
- Overall, all models achieved an accuracy score of 70% and above.

Business / Clinical KPI Success Criteria:

The model demonstrates high recall for malignant skin lesions, supporting earlier identification of potentially life-threatening skin cancers.



1. Extremely skewed dataset.
2. Model takes a long time to train.
3. Most images look quite similar.
4. Model needs more varied images.

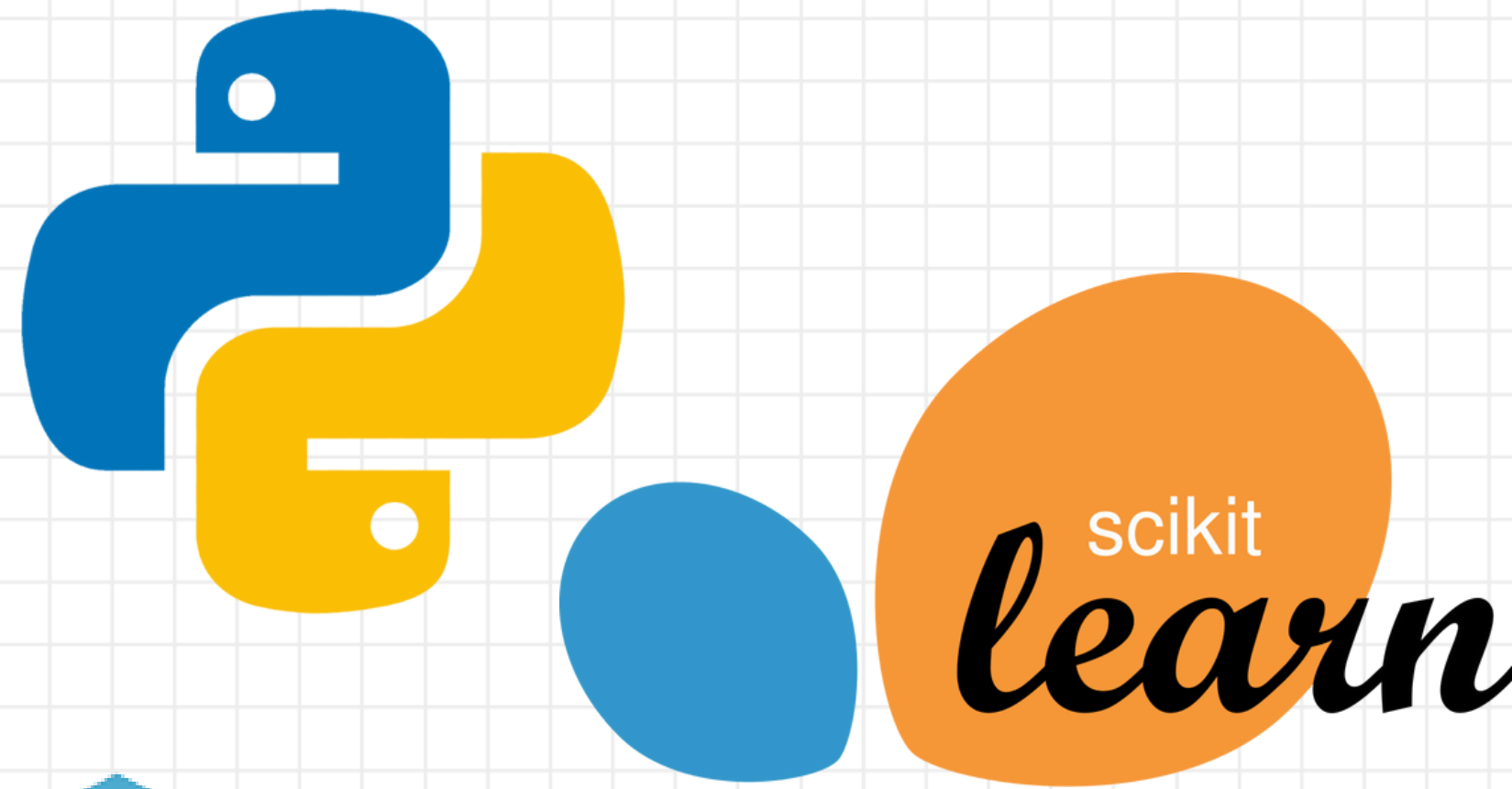


Challenges and Limitations

Future Work and Recommendations

- Run more epochs to improve accuracy
- Less skewed data.
- Experiment with hyperparameter tuning to avoid the effects of class imbalance





Tech Stack

- Language: Python
- Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Tensorflow, MobileNetV2
- Infrastructure: Streamlit

**Thank
you!**

